

第二章 电磁波的传播和传输

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北京理工大学

Beijing Institute of Technology

- 平面波解
- TEM波-横电磁波
- 波的极化
- 有耗媒质中的电磁波传播
- 群速
- 坡印廷定律

波印亭定律

用 $\mathbf{E}$ 点乘安培定律，用 $\mathbf{H}$ 点乘法拉第定律

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}) - \mathbf{H} \cdot (\nabla \times \mathbf{E}) = \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} + \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} + \mathbf{E} \cdot \mathbf{J}$$

利用

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot \nabla \times \mathbf{E} - \mathbf{E} \cdot \nabla \times \mathbf{H}$$

简化

$$\mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} + \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} + \mathbf{E} \cdot \mathbf{J} = -\nabla \cdot (\mathbf{E} \times \mathbf{H})$$

坡印亭定律

坡印亭矢量(Poynting vector)

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} = \text{Re}(\tilde{\mathbf{E}}) \times \text{Re}(\tilde{\mathbf{H}}) \neq \text{Re}(\tilde{\mathbf{E}} \times \tilde{\mathbf{H}})$$

波印亭定律

假设沿 $\hat{z}$ 方向传播的平面波的电场为

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}}E_0 e^{-(\alpha+j\beta)z} \quad (2.1.63a)$$

$$\eta = |\eta| e^{j\theta_\eta} \quad \tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{E_0}{\eta} e^{-(\alpha+j\beta)z} = \hat{\mathbf{y}} \frac{E_0}{|\eta|} e^{-(\alpha+j\beta)z - j\theta_\eta} \quad (2.1.63b)$$

$$\begin{aligned} \mathbf{S}(z, t) &= \mathbf{E}(z, t) \times \mathbf{H}(z, t) \\ &= \operatorname{Re}(\tilde{\mathbf{E}}(z) e^{j\omega t}) \times \operatorname{Re}(\tilde{\mathbf{H}}(z) e^{j\omega t}) \\ &= \hat{\mathbf{z}} \frac{E_0^2}{|\eta|} e^{-2\alpha z} \cos(\omega t - \beta z) \cos(\omega t - \beta z - \theta_\eta) \\ &= \hat{\mathbf{z}} \frac{E_0^2}{2|\eta|} e^{-2\alpha z} \left[\cos \theta_\eta + \cos(2\omega t - 2\beta z - \theta_\eta) \right] \end{aligned} \quad (2.1.64)$$

波印亭定律

一个周期内的平均能流密度矢量

$$\mathbf{S}_{av} = \frac{1}{T} \int_0^T \mathbf{S} dt = \frac{\omega}{2\pi} \int_0^T \mathbf{S} dt \quad (2.1.65)$$

把(2.1.64)式代入(2.1.65)

$$\mathbf{S}_{av} = \frac{1}{T} \int_0^T \mathbf{S} dt = \hat{\mathbf{z}} \frac{E_0^2}{2|\eta|} e^{-2\alpha z} \cos \theta_\eta \quad (\text{W/m}^2) \quad (2.1.66)$$

利用复矢量，我们还可以更简便地表述 $\mathbf{S}_{av}$

波印亭定律

考虑到

$$\operatorname{Re}(\tilde{A}) = \frac{1}{2}(\tilde{A} + \tilde{A}^*)$$

$$\mathbf{E} \triangleq \tilde{\mathbf{E}}e^{j\omega t} \quad \mathbf{H} \triangleq \tilde{\mathbf{H}}e^{j\omega t}$$

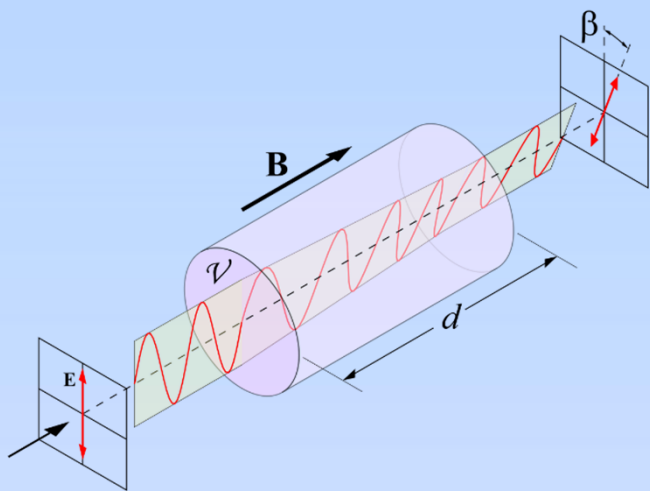
$$\begin{aligned} \mathbf{S}(t) &= \operatorname{Re}(\mathbf{E}) \times \operatorname{Re}(\mathbf{H}) \\ &= \frac{1}{2}(\mathbf{E} + \mathbf{E}^*) \times \frac{1}{2}(\mathbf{H} + \mathbf{H}^*) \end{aligned} \quad (2.1.67)$$

$$\begin{aligned} &= \frac{1}{4}(\mathbf{E} \times \mathbf{H} + \mathbf{E}^* \times \mathbf{H}^* + \mathbf{E}^* \times \mathbf{H} + \mathbf{E} \times \mathbf{H}^*) \\ &= \frac{1}{2}\operatorname{Re}(\tilde{\mathbf{E}} \times \tilde{\mathbf{H}} + \tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*) \end{aligned}$$

$$\mathbf{S}_{av} = \frac{1}{T} \int_0^T \mathbf{S} dt = \frac{1}{T} \int_0^T \frac{1}{2} \operatorname{Re}(\mathbf{E} \times \mathbf{H}^*) dt = \frac{1}{2} \operatorname{Re}(\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*) \quad (\text{W/m}^2) \quad (2.1.68)$$

时谐场平均坡印亭矢量的计算式

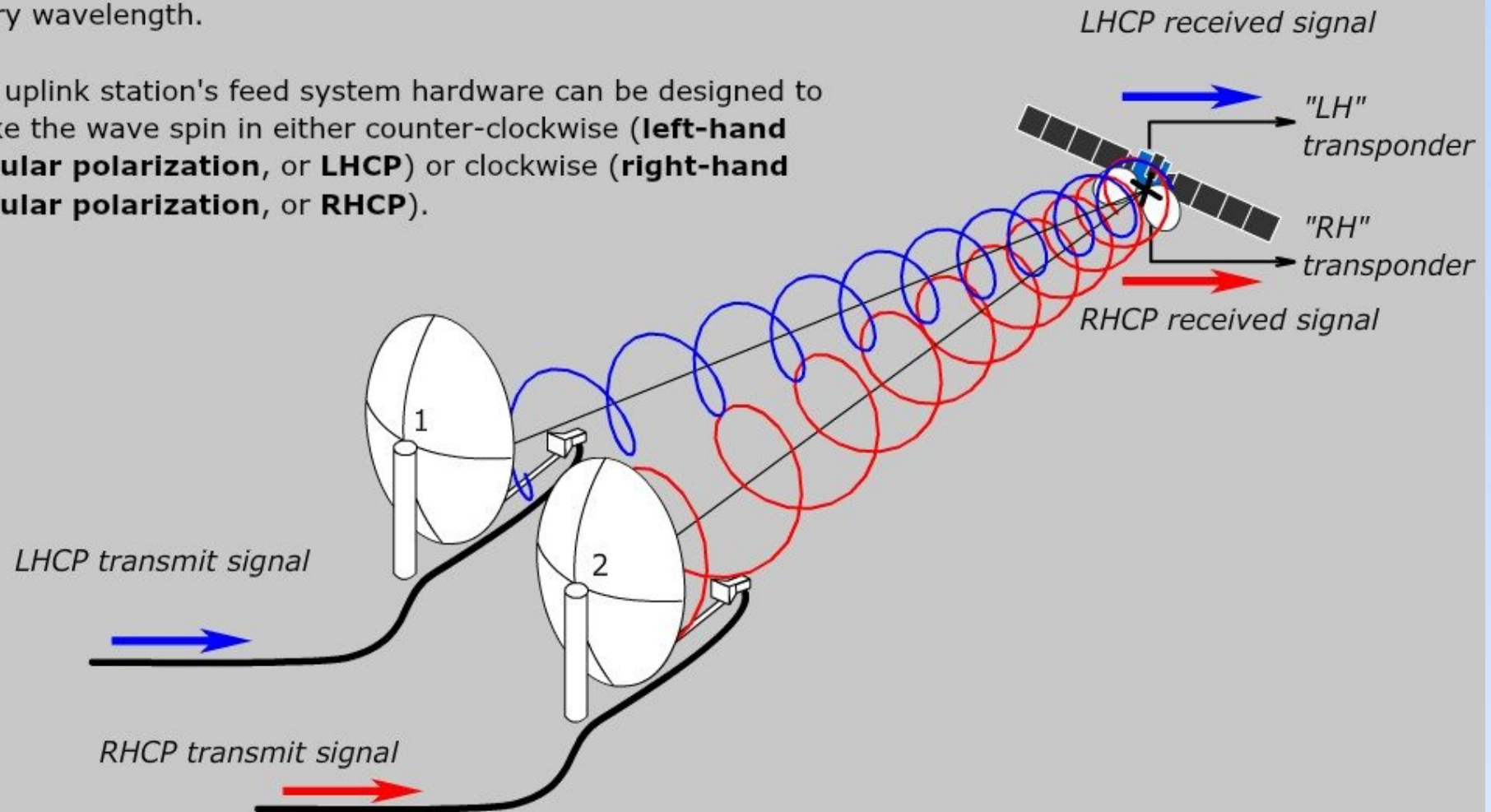
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every wavelength.

The uplink station's feed system hardware can be designed to make the wave spin in either counter-clockwise (**left-hand circular polarization, or LHCP**) or clockwise (**right-hand circular polarization, or RHCP**).



电磁波传播

- 无限大均匀媒质中的传播
- 层状媒质中电磁波的传播

■ 层状媒质中电磁波的传播

➤ 层状媒质模型

- 应用背景（微带、集成射频电路、半空间）

➤ TE、TM波

- 两组独立的麦克斯韦方程组的解（前提条件）
- 物理概念（顾名思义）
- 传输线网络方程（坐标变换与空域-谱域变换）

➤ 反射与透射系数

- 场模型（边界连续性条件）
- 路模型（阻抗失配）

➤ 全折射

- 非磁材料、TM波

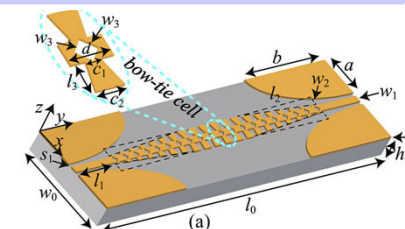
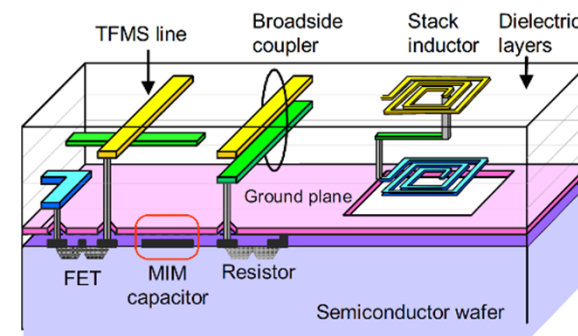
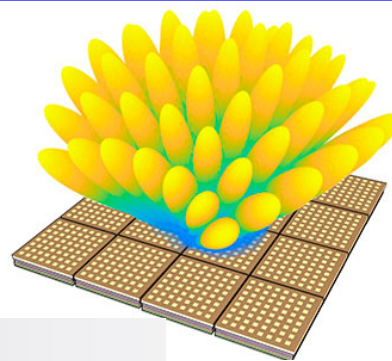
➤ 全反射

- 消逝波

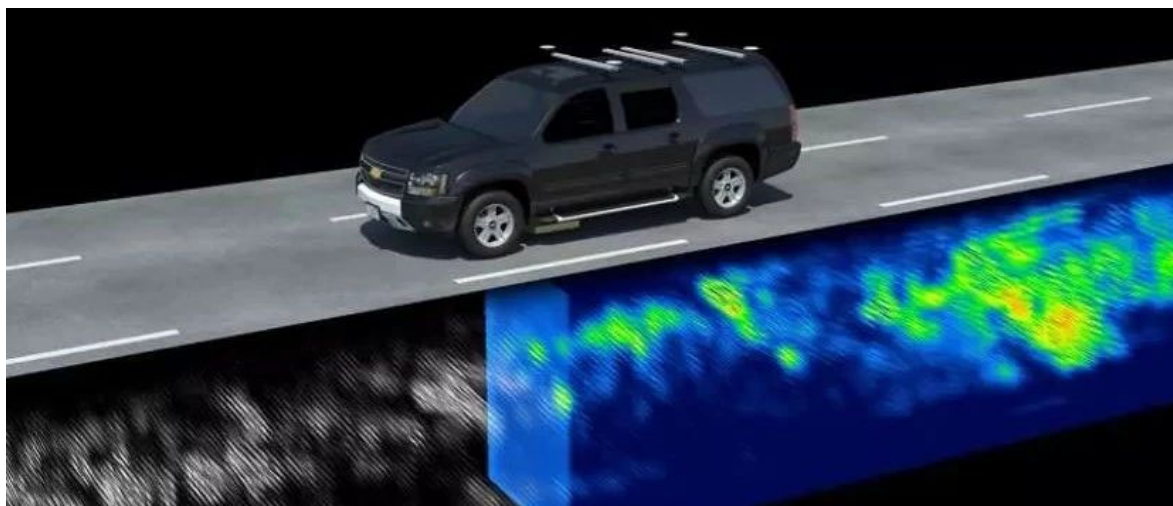
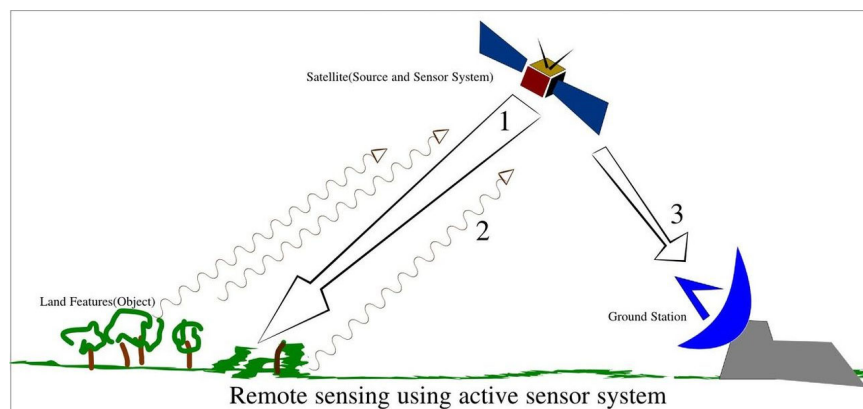
层状媒质中电磁波的传播

微带天线、电路

- 天线
- 滤波器
- 耦合器



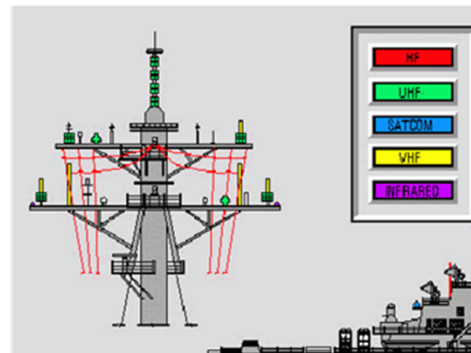
探地雷达、遥感...



层状媒质中电磁波的传播

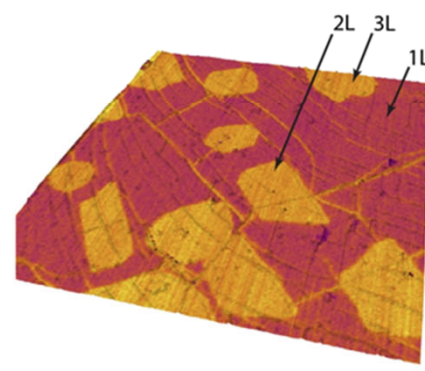
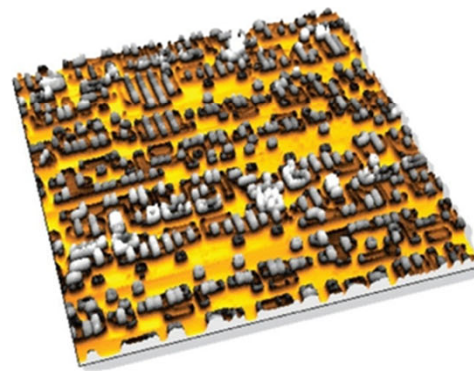
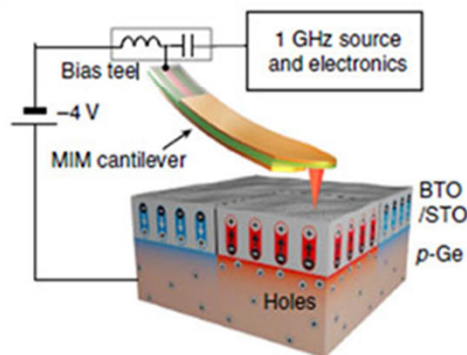
海面背景问题

- 隐身：电大、多尺度
- 天线：电磁兼容
- 探测：(准)周期结构
- 对抗：复合材料



前沿问题

- 微纳探测
- 低维、超材料



层状媒质中电磁波的传播

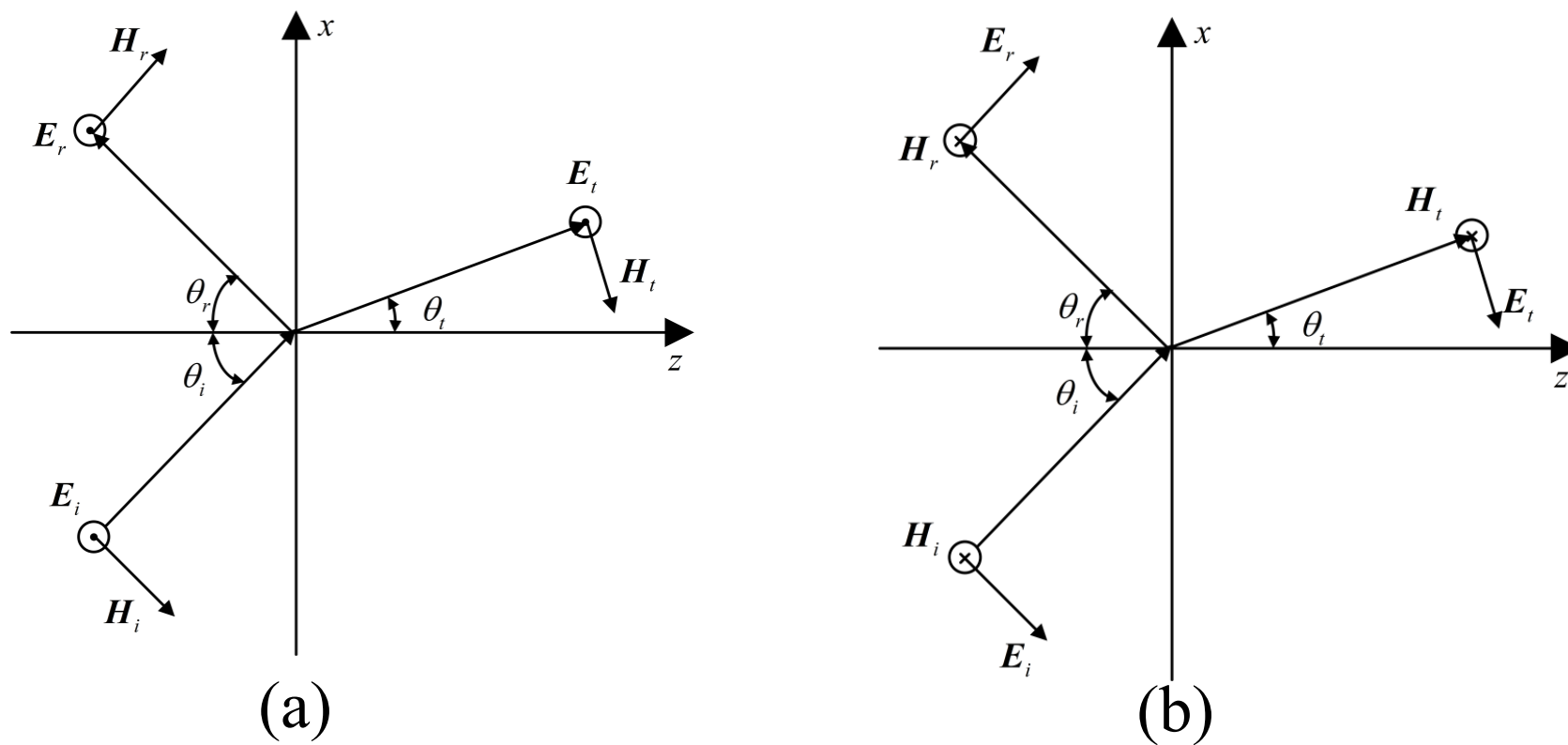


图2-3 平面波从媒质1入射到媒质2 (a)TE波 (b)TM波

Maxwell方程组

➤ Maxwell's Equations

$$\begin{aligned}\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{H} &= \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J} \\ \nabla \cdot \mathbf{D} &= \rho_{total} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \cdot \mathbf{J} &= -\frac{\partial \rho_{total}}{\partial t}\end{aligned}$$

Time domain

$e^{j\omega t}$
➔

$$\begin{aligned}\nabla \times \mathbf{E} &= -j\omega\mu\mathbf{H} \\ \nabla \times \mathbf{H} &= -j\omega\epsilon\mathbf{E} + \mathbf{J} \\ \nabla \cdot \mathbf{D} &= \rho_{total} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \cdot \mathbf{J} &= -j\omega\rho_{total}\end{aligned}$$

Frequency domain

Spatial Domain

2D-FFT
➔

$$\begin{aligned}\tilde{\nabla} \times \tilde{\mathbf{E}} &= -j\omega\mu\tilde{\mathbf{H}} \\ \tilde{\nabla} \times \tilde{\mathbf{H}} &= -j\omega\epsilon\tilde{\mathbf{E}} + \tilde{\mathbf{J}} \\ \tilde{\nabla} \cdot \tilde{\mathbf{D}} &= \tilde{\rho}_{total} \\ \tilde{\nabla} \cdot \tilde{\mathbf{B}} &= 0 \\ \tilde{\nabla} \cdot \tilde{\mathbf{J}} &= -j\omega\tilde{\rho}_{total}\end{aligned}$$

Spectral domain

谱域Maxwell方程组

$$\tilde{\nabla} \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}} - \tilde{\mathbf{M}}$$

$$\tilde{\nabla} \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}} + \tilde{\mathbf{J}}$$

$$\tilde{\nabla} = -jk_x\hat{x} - jk_y\hat{y} + \frac{\partial}{\partial z}\hat{z}$$

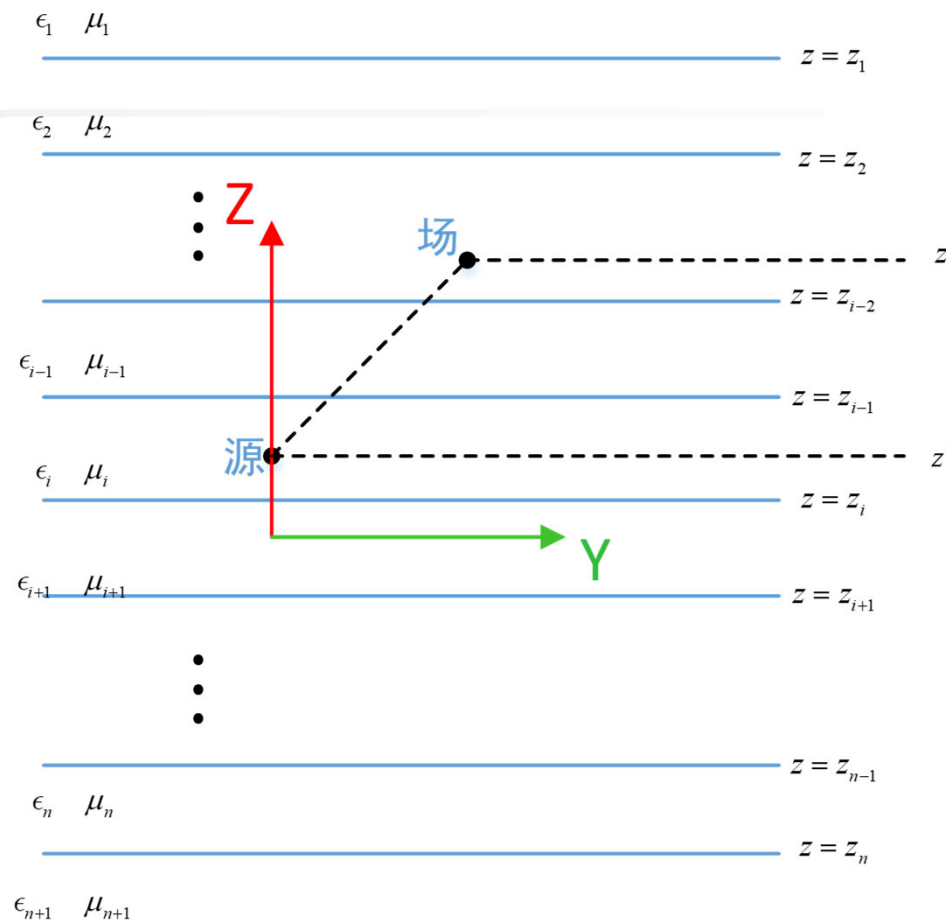
坐标变换:

$$k_x = k_\rho \cos \phi \quad k_y = k_\rho \sin \phi$$

$$k_\rho = \sqrt{k_x^2 + k_y^2} = \sqrt{k^2 - k_z^2} \quad \phi = \arctan(k_y / k_x)$$

$$\tilde{U}_x = \tilde{U}_\rho \cos \phi - \tilde{U}_\phi \sin \phi \quad \tilde{U}_y = \tilde{U}_\rho \sin \phi + \tilde{U}_\phi \cos \phi$$

$$\tilde{U}_\rho = \tilde{U}_x \cos \phi + \tilde{U}_y \sin \phi \quad \tilde{U}_\phi = -\tilde{U}_x \sin \phi + \tilde{U}_y \cos \phi$$



$$\cos \phi \left[\frac{\partial \tilde{E}_\phi}{\partial z} - j\omega\mu \tilde{H}_\rho - \tilde{M}_\rho \right] + \sin \phi \left[\frac{\partial \tilde{E}_\rho}{\partial z} + jk_\rho \tilde{E}_z + j\omega\mu \tilde{H}_\phi + \tilde{M}_\phi \right] = 0$$

$$\cos \phi \left[\frac{\partial \tilde{E}_\rho}{\partial z} + jk_\rho \tilde{E}_z + j\omega\mu \tilde{H}_\phi + \tilde{M}_\phi \right] + \sin \phi \left[-\frac{\partial \tilde{E}_\phi}{\partial z} + j\omega\mu \tilde{H}_\rho + \tilde{M}_\rho \right] = 0$$

$$-jk_\rho \tilde{E}_\phi + j\omega\mu \tilde{H}_z + \tilde{M}_z = 0$$

$$\cos \phi \left[-\frac{\partial \tilde{H}_\phi}{\partial z} - j\omega\epsilon \tilde{E}_\rho - \tilde{J}_\rho \right] + \sin \phi \left[-\frac{\partial \tilde{H}_\rho}{\partial z} + jk_\rho \tilde{H}_z + j\omega\epsilon \tilde{E}_\phi + \tilde{J}_\phi \right] = 0$$

$$\cos \phi \left[-\frac{\partial \tilde{H}_\rho}{\partial z} - jk_\rho \tilde{H}_z + j\omega\epsilon \tilde{E}_\phi + \tilde{J}_\phi \right] + \sin \phi \left[\frac{\partial \tilde{H}_\phi}{\partial z} + j\omega\epsilon \tilde{E}_\rho + \tilde{J}_\rho \right] = 0$$

$$jk_\rho \tilde{H}_\phi + j\omega\epsilon \tilde{E}_z + \tilde{J}_z = 0$$

TE 波

- 源: $\tilde{J}_\phi$ $\tilde{M}_\rho$ $\tilde{M}_z$
- 场: $\tilde{E}_\phi$ $\tilde{H}_\rho$ $\tilde{H}_z$

$$\frac{\partial \tilde{E}_\phi}{\partial z} = j\omega\mu \tilde{H}_\rho + \tilde{M}_\rho$$

$$\frac{\partial \tilde{H}_\rho}{\partial z} = \left(j\omega\epsilon + \frac{k_\rho^2}{j\omega\mu} \right) \tilde{E}_\rho + \tilde{J}_\phi + \frac{k_\rho}{\omega\mu} \tilde{M}_z$$

$$\tilde{H}_z = \frac{k_\rho}{\omega\mu} \tilde{E}_\phi - \frac{1}{j\omega\mu} \tilde{M}_z$$

TM 波

- 源: $\tilde{M}_\phi$ $\tilde{J}_\rho$ $\tilde{J}_z$
- 场: $\tilde{H}_\phi$ $\tilde{E}_\rho$ $\tilde{E}_z$

$$\frac{\partial \tilde{H}_\phi}{\partial z} = -j\omega\epsilon \tilde{H}_\rho - \tilde{J}_\rho$$

$$\frac{\partial \tilde{E}_\rho}{\partial z} = -\left(j\omega\mu + \frac{k_\rho^2}{j\omega\epsilon} \right) \tilde{H}_\phi - \tilde{M}_\phi + \frac{k_\rho}{\omega\epsilon} \tilde{J}_z$$

$$\tilde{E}_z = -\frac{k_\rho}{\omega\epsilon} \tilde{H}_\phi - \frac{1}{j\omega\epsilon} \tilde{J}_z$$

传输线网络方程

$$\frac{\partial V(z)}{\partial z} = -(R + j\omega L)I(z) + V_0\delta(z - z')$$

$$\frac{\partial I(z)}{\partial z} = -(G + j\omega C)V(z) - I_0\delta(z - z')$$

TE波 $V_0^h = -\tilde{M}_{\rho 0}$ $I_0^h = -\tilde{J}_{\phi 0} - \frac{k_\rho}{\omega\mu} \tilde{M}_{z0}$

TM波 $V_0^e = -\tilde{M}_{\phi 0} + \frac{k_\rho}{\omega\mu} \tilde{J}_{z0}$ $I_0^e = \tilde{J}_{\rho 0}$

层状媒质中电磁波的传播

TE波

$$\mathbf{E}_i = \hat{y}e^{-j(k_{ix}x+k_{iz}z)} \quad (2.1.69)$$

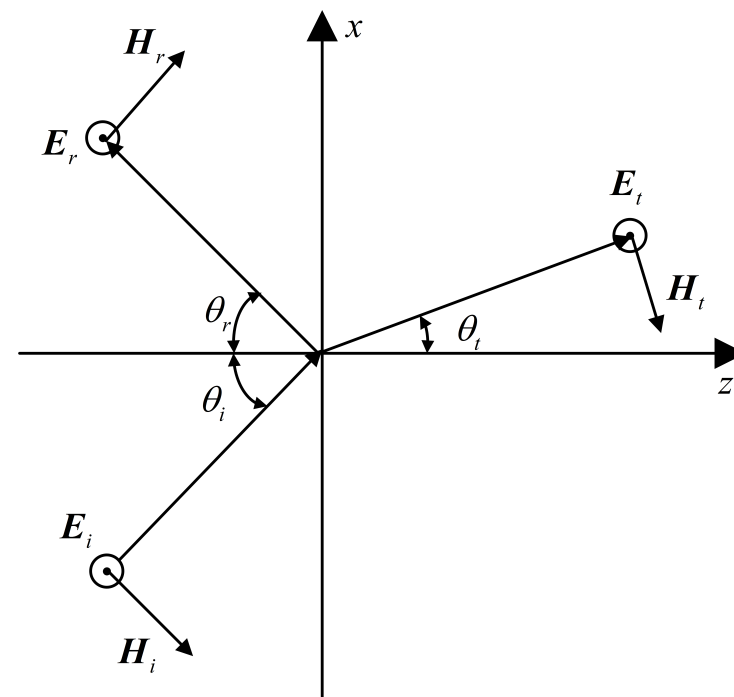
$$\mathbf{H}_i = \frac{1}{\omega\mu_1}(-\hat{x}k_{iz} + \hat{z}k_{ix})e^{-j(k_{ix}x+k_{iz}z)} \quad (2.1.70)$$

$$\mathbf{E}_r = \hat{y}Re^{-j(k_{rx}x-k_{rz}z)} \quad (2.1.71)$$

$$\mathbf{H}_r = \frac{R}{\omega\mu_1}(\hat{x}k_{rz} + \hat{z}k_{rx})e^{-j(k_{rx}x-k_{rz}z)} \quad (2.1.72)$$

$$\mathbf{E}_t = \hat{y}Te^{-j(k_{tx}x+k_{tz}z)} \quad (2.1.73)$$

$$\mathbf{H}_t = \frac{T}{\omega\mu_2}(-\hat{x}k_{tz} + \hat{z}k_{tx})e^{-j(k_{tx}x+k_{tz}z)} \quad (2.1.74)$$



(a)

图2-3 平面波从媒质1
入射到媒质2 (a)TE波

层状媒质中电磁波的传播

在两种媒质中分别有下列色散关系

$$k_{ix}^2 + k_{iz}^2 = \omega^2 \mu_1 \varepsilon_1 \quad (2.1.75a)$$

$$k_{rx}^2 + k_{rz}^2 = \omega^2 \mu_1 \varepsilon_1 \quad (2.1.75b)$$

$$k_{tx}^2 + k_{tz}^2 = \omega^2 \mu_2 \varepsilon_2 \quad (2.1.75c)$$

且由交界面切向电场和磁场相等得

$$e^{-jk_{ix}x} + Re^{-jk_{rx}x} = Te^{-jk_{tx}x} \quad (2.1.76)$$

$$-\frac{k_{iz}}{\mu_1} e^{-jk_{ix}x} + \frac{Rk_{rz}}{\mu_1} e^{-jk_{rx}x} = -\frac{Tk_{tz}}{\mu_2} e^{-jk_{tx}x} \quad (2.1.77)$$

相位匹配条件 $k_{ix} = k_{rx} = k_{tx}$ (2.1.78)

由(2.1.75a)和(2.1.75b)可知 $k_{iz} = k_{rz}$

层状媒质中电磁波的传播

例题2.3

将(2.1.78)代入(2.1.76)和(2.1.77)得

$$1 + R = T \quad (2.1.79)$$

$$k_z (1 - R) = \frac{\mu_1}{\mu_2} T k_{tz} \quad (2.1.80)$$

由(2.1.79)和(2.1.80)式可解出

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} \quad (2.1.81) \quad T = \frac{2Z_2}{Z_2 + Z_1} \quad (2.1.82)$$

$$\text{其中 } Z_1 = \frac{\omega \mu_1}{k_z} \quad (2.1.83a) \quad Z_2 = \frac{\omega \mu_2}{k_{tz}} \quad (2.1.83b)$$

对于TM波，用同样方式可以得到相同表达式(2.1.79)，(2.1.82)，(2.1.83)；只不过，对于TM波

$$Z_1 = \frac{k_z}{\omega \varepsilon_1} \quad (2.1.84a) \quad Z_2 = \frac{k_{tz}}{\omega \varepsilon_2} \quad (2.1.84b)$$

层状媒质中电磁波的传播

由(2.1.78)可推出下面入射角 θ_i ,反射角 θ_r ,以及折射角 θ_t 的关系

$$\theta_i = \theta_r \quad (2.1.85a) \quad n_1 \sin \theta_i = n_2 \sin \theta_t \quad (2.1.85b)$$

媒质1, 2的折射率 $n_1 = \sqrt{\mu_1 \epsilon_1}, n_2 = \sqrt{\mu_2 \epsilon_2}$

斯涅尔定律(Snell's law)

当 $n_1 > n_2$ 时, 如果入射角 θ_i 大于下面临界角

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) \quad (2.1.86)$$

不论是TE波, 还是TM波, 都将发生全反射 (**total reflection**)

层状媒质中电磁波的传播

由(2.1.99)式可知，在 $Z_t = Z$ 时，发生零反射，即全折射（**total refraction**）

对于TE波

$$\frac{\mu_1}{n_1 \cos \theta_i} = \frac{\mu_2}{n_2 \cos \theta_t} \quad (2.1.87)$$

对于自然界中常见的电介质媒质(即 $\mu_1 = \mu_2 = \mu_0$)

$$n_1 \cos \theta_i = n_2 \cos \theta_t \quad (2.1.88)$$

在 $n_1 \neq n_2$ 时，(2.1.88)式显然与(2.1.85b)式矛盾。

这就说明TE波不可能发生全折射

层状媒质中电磁波的传播

对于TM波

$$\frac{\cos \theta_i}{n_1} = \frac{\cos \theta_t}{n_2} \quad (2.1.89)$$

将(2.1.107)与(2.1.103b)相乘得

$$\sin 2\theta_i = \sin 2\theta_t \quad (2.1.90)$$

两个解： $\theta_i = \theta_t$ ； 在 $n_1 \neq n_2$ 时， 与(2.1.85b)式矛盾

$\theta_i = \pi/2 - \theta_t$ ； 代入(2.1.85b)， 求出发生全折射的入

射角为 $\theta_B = \tan^{-1} \left(\frac{n_2}{n_1} \right) \quad (2.1.91)$

布鲁斯特角(Brewster Angle)

层状媒质中电磁波的传播

空气入射到海水情况下，TE和TM波反射系数随入射角度的变化曲线

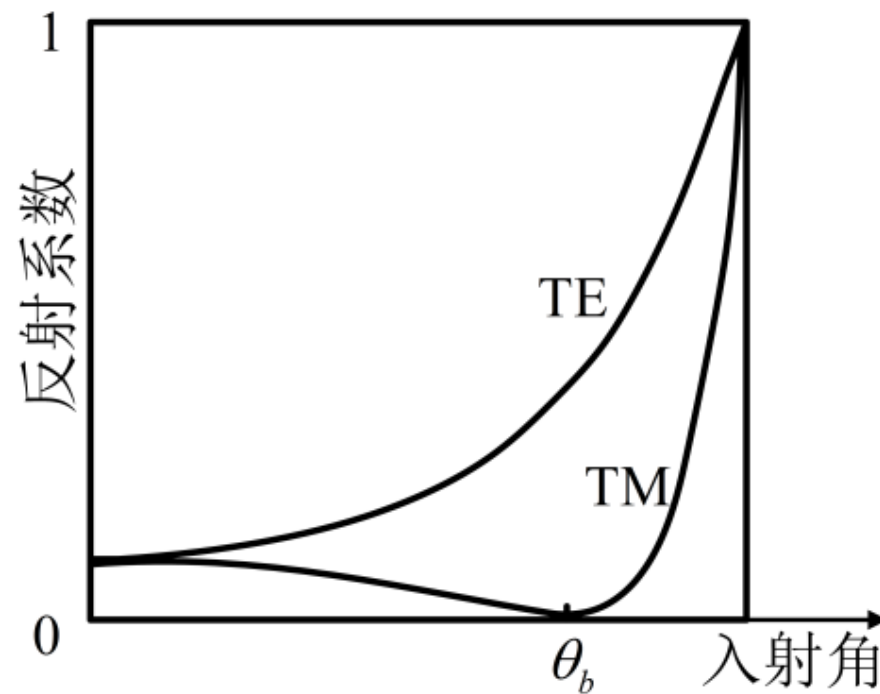
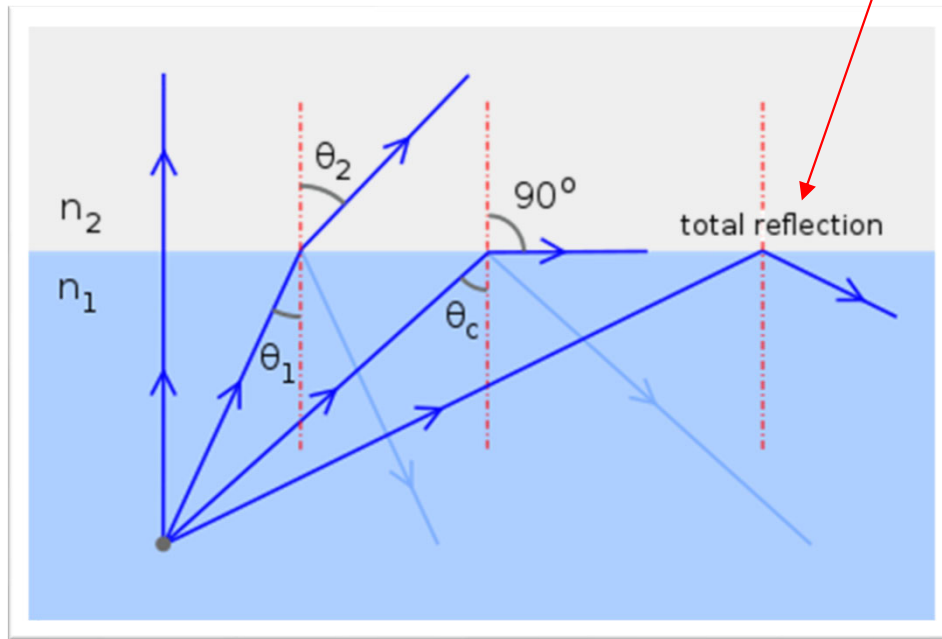


图2-4 TE和TM波的反射系数

层状媒质中电磁波的传播

从玻璃照射空气情况下，TE和TM波都会发射全反射现象

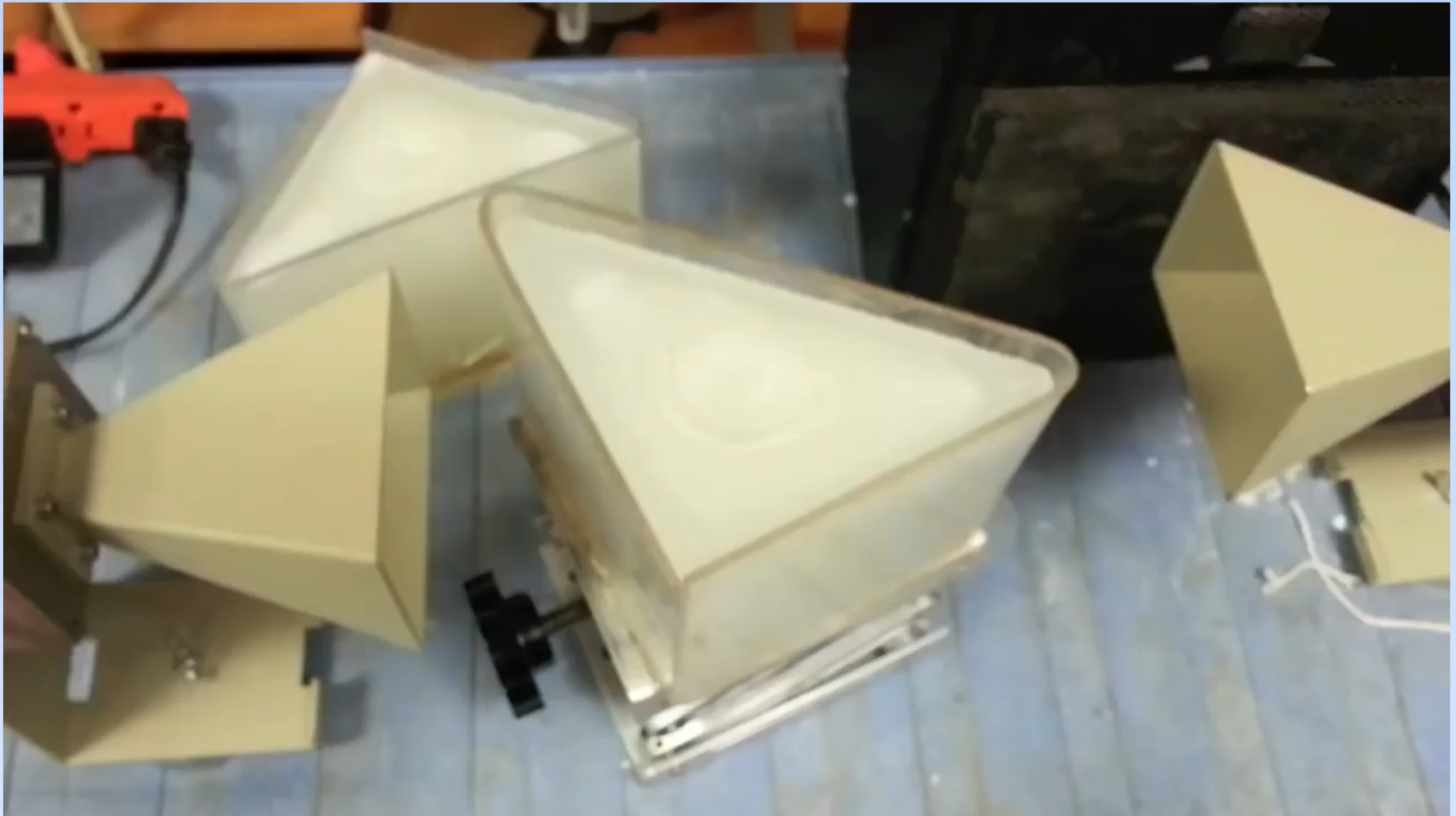


全反射时另一侧的场分布如何？

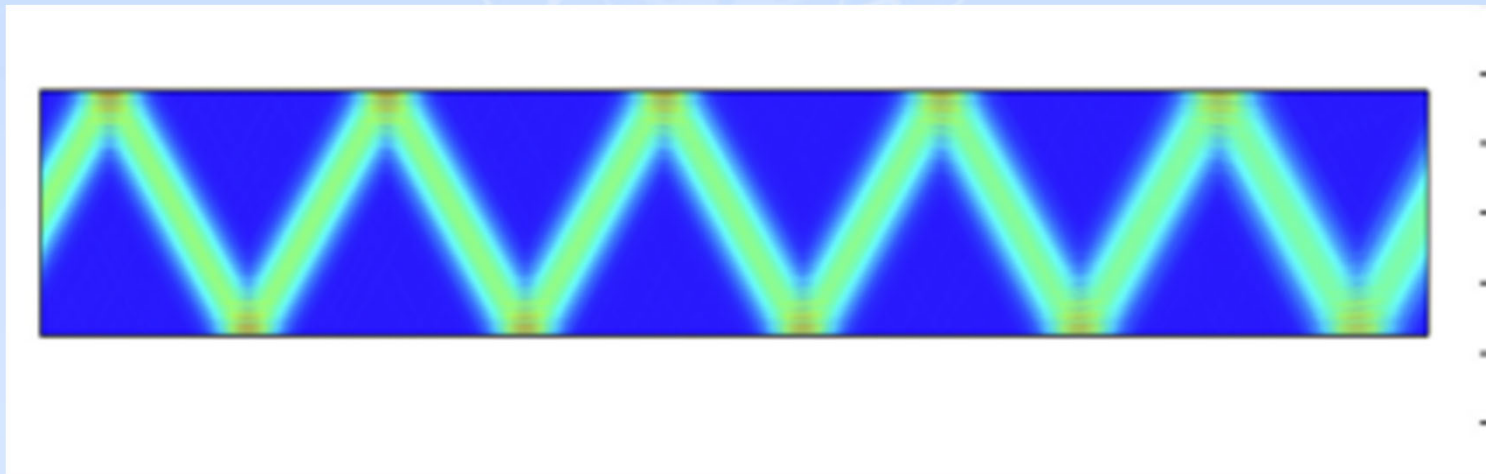
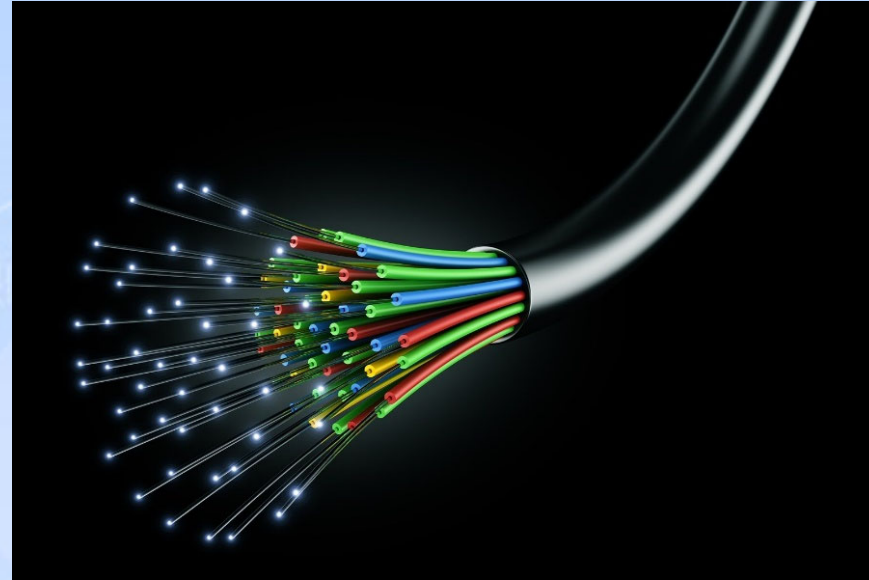
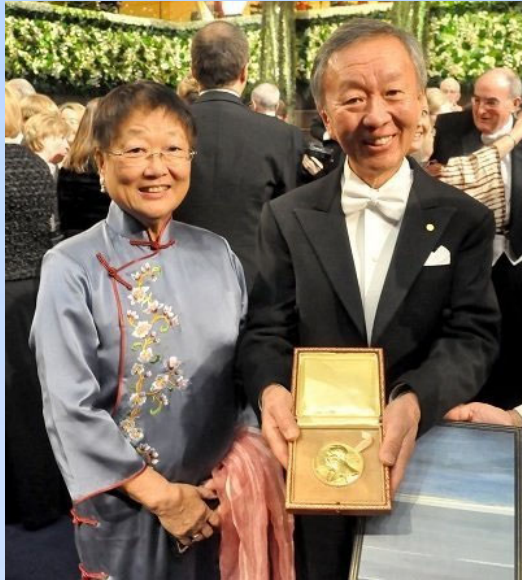
A 存在场分布

B 不存在场分布

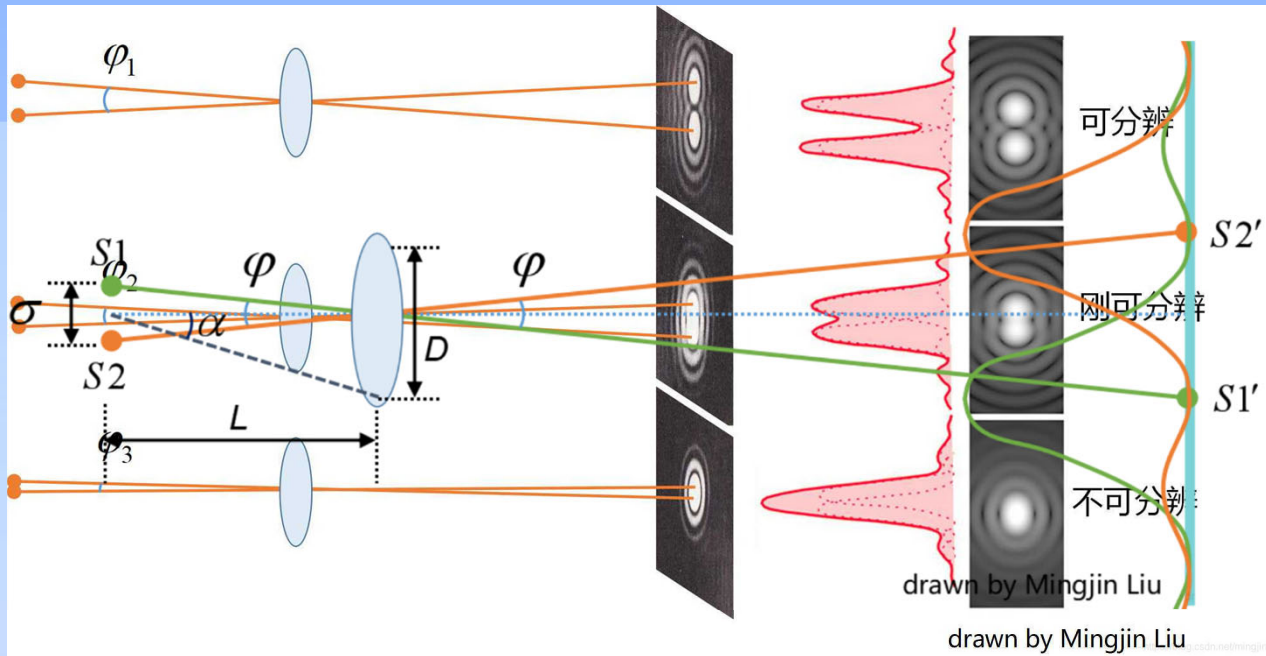
层状媒质中电磁波的传播



层状媒质中电磁波的传播



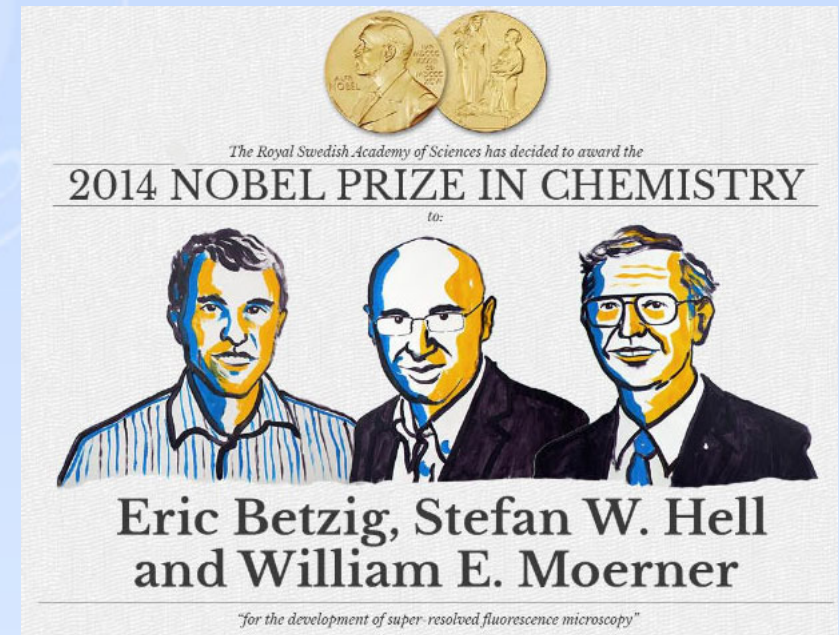
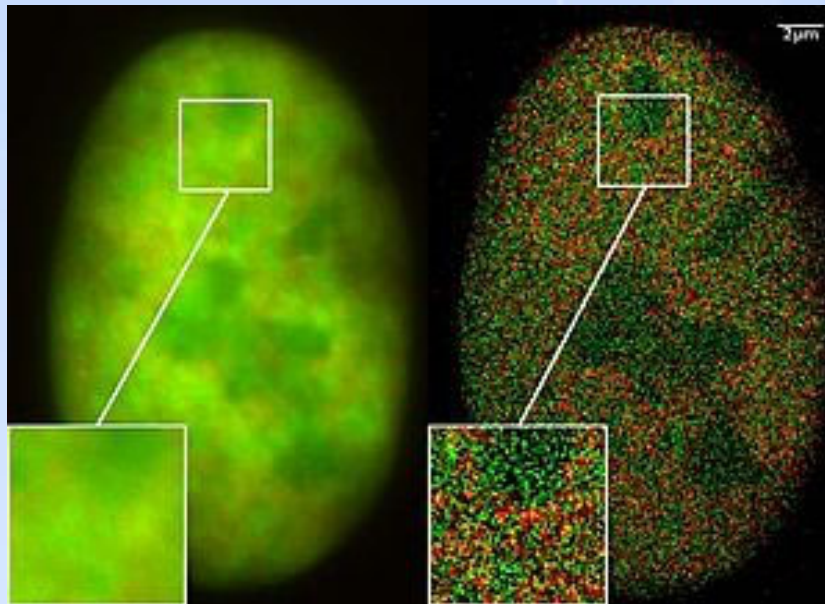
K. C. Kao, G. A. Hockham (1966), "Dielectric-fibre surface waveguides for optical frequencies" (《光频率介质纤维表面波导》), *Proc. IEE* **113** (7): 1151–1158.



$$\phi = \frac{1.22\lambda}{D}$$

$$\sigma = \phi L = \frac{1.22\lambda f}{D} = \frac{0.61\lambda}{NA}$$

$$NA = n \sin \alpha$$



Betzig, Eric, and Robert J. Chichester. "Single molecules observed by near-field scanning optical microscopy." *Science* 262.5138 (1993): 1422-1425.

例题2.3

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从振幅和坡印亭矢量角度分析**TE**波正入射到两介质交界面处的情况。

例题2.3

解：

$$\text{正入射时} \quad k_x = k_{rx} = 0, \quad k_z = k_{rz} = k_1 \quad (\text{e2.4.1})$$

介质1中电场和磁场

$$\mathbf{E}_1 = \mathbf{E}_i + \mathbf{E}_r = \hat{y}(e^{-jk_1z} + Re^{jk_1z}) = \hat{y}[(1+R)e^{-jk_1z} + j2R\sin(k_1z)] \quad (\text{e2.4.2})$$

$$\mathbf{H}_1 = \mathbf{H}_i + \mathbf{H}_r = -\hat{x}\frac{k_z}{\omega\mu_1}(e^{-jk_1z} - Re^{jk_1z}) = -\hat{x}\frac{k_z}{\omega\mu_1}[(1+R)e^{-jk_1z} - j2R\sin(k_1z)] \quad (\text{e2.4.3})$$

合成波电场的振幅为

$$\begin{aligned} |\mathbf{E}_1| &= |e^{-jk_zz} + Re^{jk_zz}| = |1 + Re^{j2k_zz}| \\ &= |1 + R\cos(2k_zz) + jR\sin(2k_zz)| \\ &= \sqrt{1 + R^2 + 2R\cos(2k_zz)} \end{aligned} \quad (\text{e2.4.4})$$

例题2.3

介质1, 2都为理想介质, Z_1, Z_2, R, T 都为实数

当 $Z_2 > Z_1$ 时, $R > 0$

电场在 $2k_1z = -2n\pi$ ($n = 0, 1, 2, \dots$) 即 $z = -\frac{n\lambda_1}{2}$ 处取得最大振幅

$$|E_1|_{\max} = 1 + R \quad (\text{e2.4.5})$$

同理, 在 $z = -\frac{(2n+1)\lambda_1}{4}$ 处取得最小振幅值

$$|E_1|_{\min} = 1 - R \quad (\text{e2.4.6})$$

在两媒质交界面上

$$|E_1(0^-)| = |E_1|_{\max} = 1 + R = T = |E_2(0^+)| \quad (\text{e2.4.7})$$

$Z_2 < Z_1$ 情况同理讨论不赘述

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媒质1中的平均功率密度为

$$\begin{aligned} \mathbf{S}_{1av} &= \frac{1}{2} \text{Re} [\hat{y} E_1 \times (-\hat{x}) H_1^*] = \hat{z} \frac{1}{2} \text{Re} \left\{ \left(e^{-jk_1 z} + R e^{jk_1 z} \right) \frac{k_1}{\omega \mu_1} \left(e^{-jk_1 z} - R e^{jk_1 z} \right)^* \right\} \quad (\text{e2.4.8}) \\ &= \hat{z} \frac{k_1}{2\omega \mu_1} \text{Re} \left[\left(1 + R e^{j2k_1 z} \right) \left(1 - R e^{-j2k_1 z} \right) \right] = \hat{z} \frac{k_1}{2\omega \mu_1} \text{Re} \left[\left(1 - R^2 \right) + j2R \sin(2k_1 z) \right] \\ &= \hat{z} \frac{k_1}{2\omega \mu_1} (1 - R^2) \end{aligned}$$

媒质2中的平均功率密度为

$$\mathbf{S}_{2av} = \frac{1}{2} \text{Re} [\hat{y} E_2 \times (-\hat{x}) H_2^*] = \hat{z} \frac{1}{2} \text{Re} \left\{ T e^{-jk_2 z} \left(\frac{T k_{tz}}{\omega \mu_2} e^{-jk_2 z} \right)^* \right\} = \hat{z} \frac{T^2 k_{tz}}{2\omega \mu_2} \quad (\text{e2.4.9})$$

$$\mathbf{S}_{1av} = \frac{k_1}{\omega \mu_1} (1 - R^2) = \frac{k_1}{\omega \mu_1} (1 - R) \cdot (1 + R) = \frac{T k_{tz}}{\omega \mu_2} \cdot T = \frac{T^2 k_{tz}}{\omega \mu_2} = \mathbf{S}_{2av} \quad (\text{e2.4.10})$$

可见，从媒质1到媒质2，波的坡印亭矢量，即功率密度、功率也是连续的

例题2.5

例题2.5 设平面波从空气一侧以倾斜角 θ 入射到铜(电导率 $\sigma = 5.8 \times 10^7 \text{ S/m}$)的界面。求 $f = 10\text{GHz}$ 频率下此界面的反射波。

解：

对于一般金属来说

$$\varepsilon' \approx 1, \quad \varepsilon'' = \frac{\sigma}{\omega \varepsilon_0} \quad (\text{e2.5.1})$$

在微波频率， $\varepsilon'' \gg \varepsilon'$ 。此时无论TE波还是TM波纵向传播系数

$$k_{tz} = \sqrt{k_0^2 (\varepsilon' - j\varepsilon'') - k_x^2} = \sqrt{-jk_0^2 \varepsilon''} = \sqrt{\frac{\omega \mu_0 \sigma}{2}} (1 - j) = \frac{1}{\delta} (1 - j) \quad (\text{e2.5.2})$$

$$\delta = \sqrt{\frac{2}{\omega \mu_0 \sigma}} \text{ 为趋肤深度}$$

$$Z_{2TE} = Z_{2TM} = \sqrt{\frac{\omega \mu_0}{2\sigma}} (1 + j) = 2.6 \times 10^{-2} (1 + j) (\Omega) \quad (\text{e2.5.3})$$

反射系数

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} \approx -1 \quad (\text{e2.5.4})$$

在微波频段金属可视为短路，波全部反射

$$\mathbf{E}_r = -\mathbf{E}_i, \quad \mathbf{H}_r = \mathbf{H}_i \quad (\text{e2.5.5})$$